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Social Pattern Discovery for GPS Datasets

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Abstract

This work introduces a method to analyze human mobility and social interactions, highlighting the use of geohash codes for spatial clustering and the identification of social encounters with and without time constraints. Our approach uncovers “followers” by tracking how people move to subsequent geohashes within a specific time interval, revealing patterns of coordinated mobility. Our analysis includes the construction of a social graph visualizing individual connections. Additionally, we discover significant data duplication within the applied dataset (Geolife by Microsoft Research Asia), leading to a reassessment of the frequency of social interactions. This result underscores the importance of data integrity in geospatial analysis, contributing to the broader field of human mobility research by demonstrating a novel method for extracting meaningful insights from complex mobility data.

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1. Introduction

Understanding human mobility patterns and social interactions is crucial for advancements in urban planning, transportation systems, and public health. The advent of GPS (Global Positioning System) technology has enabled the collection of large-scale spatial-temporal datasets, offering unprecedented insights into human movement behavior. Among these, the Geolife dataset, collected by Microsoft Research Asia in Beijing and documented by Zheng et al. [10], stands out. Despite its broad application over the past decade, a key aspect remains unexplored until now—the presence of duplicated records and user tracks, which significantly impact the dataset’s integrity and the interpretation of social interactions.

This study confronts the challenge of duplicate data, demonstrating its existence and influence on the perceived volume of social encounters within the Geolife dataset. We introduce a novel analytical methodology that leverages geohash codes for effective spatial clustering, combined with temporal analysis in order to detect and analyze social encounters and mobility patterns among individuals. The ambitions of our approach go beyond traditional GPS coordi-

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nate comparisons, allowing us to discern between casual proximity and significant social interactions. This distinction is further refined through the identification of "followers" — individuals who, after a shared geohash encounter, continue to move in synchrony to subsequent geohashes within specific time intervals, indicating a form of coordinated movement or shared activity.

By addressing the issue of data duplication, analysis of social encounters and follower behavior, our research attempts to shed light on the actual dynamics of human mobility and interaction patterns. The discovery of data duplication not only challenges existing interpretations of the Geolife dataset but also highlights the critical importance of data integrity for accurate analysis in the area of human mobility research. Our methodological approach provides new insights into the true nature of social interactions and mobility patterns, making a contribution to the fields of urban planning, location-based services, and social network analysis. The innovation of our work lies in both the analytical techniques developed for interpreting GPS data and the critical examination of the dataset itself. This dual emphasis not only advances the state of analytical methods but also enhances the reliability of research utilizing the Geolife dataset. Hopefully, the implications of our findings will offer practical insights for the design and implementation of urban infrastructure, the development of location-based services, and the understanding of social networks.

This document is structured as follow. Section 2 reviews related work in the field of human mobility and social pattern analysis. Section 3 describes our methodology, which includes data preprocessing, the application of geohash codes for spatial clustering, and the criteria for identifying social encounters and followers. Section 4 introduces the experimental setup, including the dataset used and its preprocessing steps, particularly the removal of duplicated trajectories. Section 5 presents our findings, supported by detailed statistical analysis, histograms, and a social graph visualization. Finally, Section 6 concludes the paper, summarizing our contributions to the understanding of human mobility through the lens of the Geolife dataset.

2. Literature Review

The exploration of human mobility and social interactions using GPS data and mobile technology has evolved significantly, with contributions spanning privacy concerns, social network enhancement, and advanced analytical methodologies.

The area of privacy concerns has been approached by Gambs et al. [6], who delve into the privacy implications inherent in geolocated datasets. Their work introduces a sanitization framework aimed at mitigating inference attacks, setting a foundational approach for future privacy-preserving mobility analyses. This concern for privacy is echoed in the study by Ayalon et al. [2], which investigates the engagement and efficacy of Belgium's contact tracing app, Coronalert, among young adults. Their findings highlight a critical need for balance between user privacy and the utility of any data collection for public health.

Advancements in predictive modeling of human mobility represent another significant research direction. Corrias et al. [5] pioneer the use of General Purpose Transformers (GPT) and Graph Convolutional Networks (GCNs) in order to predict human mobility, marking a departure from traditional modeling techniques. Their work showcases the potential of leveraging spatial-temporal data structures, akin to those in natural language processing, to enhance mobility prediction accuracy, thereby contributing to the broader understanding of human movement patterns.

The identification of significant locations and patterns in individual mobility is addressed by Zhou et al. [11], who propose a methodology for mining personally important places from GPS tracks. Their approach not only advances the granularity of mobility analyses but also underscores the importance of recognizing places of personal significance beyond mere frequency of visitation. Their focus on individual mobility patterns is complemented by Schreiner's discussion [9] on the societal impact of GPS integration into mobile phones, highlighting the transformative potential of location-aware computing for personal navigation and social networking.

The intersection of mobility data and social connectivity is further explored by Quercia and Capra [8] through their "FriendSensing" framework, which recommends potential friends based on co-location data captured via mobile phones' Bluetooth technology. This innovative approach to enhancing social networks through technological means is a significant step forward in leveraging everyday technology for social connectivity. Similarly, Amir et al. [1] contribute to this domain with algorithms for efficient proximity detection among friends in mobile networks, highlighting the challenges and potential of location-based services in fostering social interactions.

A comprehensive model for understanding the interplay between human mobility and social networks is presented by Zignani et al. [12], who introduce the concept of "geo-communities" to represent the spatial and social dimensions of mobility. Their analysis of GPS data to extract social behavior and identify geo-communities offers a nuanced perspective on the complex relationship between physical movement and social interactions, enriching the discourse on the potential of location-aware data to inform communication network design and social network analysis.

Together, the above reviewed studies paint a multifaceted picture of the current state of research on human mobility, privacy, social connectivity, and the technologies that connect them. They highlight the complexity of analyzing human behavior through GPS data, the evolving landscape of privacy and data integrity concerns, and the recent methodologies and models that continue to push the boundaries of what can be understood from spatial-temporal mobility data.

Building on these works, our contribution focuses on uncovering the delicate dynamics of human meetings and "followers" within GPS datasets. By identifying and addressing the challenge of duplicated records, we introduce a novel approach for analyzing spatio-temporal data, enhancing the understanding of social patterns and mobility. Our work not only addresses a gap in the existing literature regarding data duplication but also advances methodologies for detecting social encounters, contributing to the broader field of human mobility research and offering insights into the design of future location-based services.

3. Methodology

3.1. Input data format

A GPS trajectory is a sequence of chronologically arranged locations that capture the movement of a GPS enabled device. We define GPS trajectory as follows:

Definition 1. Trajectory $\mathcal{S}$ is a sequence of n locations $\{\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \dots, \mathcal{L}_n\}$, where location $\mathcal{L}$ is a tuple $(lat, lon, time, id, uid)$ such that

- *lat* is latitude in decimal degrees (e.g. 48.1518568),
- *lon* is longitude in decimal degrees (e.g. 17.0711559),
- *time* is time in seconds from 1.1.1970 (UTC),
- *id* is unique identifier of trajectory,
- *uid* is unique identifier of user,

and for $\mathcal{L}_i = (lat_i, lon_i, time_i, id_i, uid_i)$ the following holds:

$$\forall \mathcal{L}_i, \mathcal{L}_{i+1} : time_i < time_{i+1}.$$

Our proposed algorithm is capable of processing any data set that can be converted to the CSV file format with the following structure based on previously defined definition.

latitude	longitude	time	id	uid
48.1518568	17.0711559	1677749039	1	10
48.1618568	17.0721559	1677749046	1	10
...				
48.1718568	17.0731559	1677734124	2	10
....				

3.2. Data preprocessing

With the dataset structure defined and data loaded, we proceeded to calculate geohash codes for each GPS coordinate. Geohash [7] is a hierarchical spatial data structure that converts GPS locations into a unique string of letters and digits. This encoding process involves dividing space into grid-shaped buckets using a Z-order curve and applying base32 encoding to the latitude and longitude values. Geohashes with a common prefix indicate spatial proximity of

their corresponding points, although it is possible for points to be spatially close without sharing a prefix. By converting each GPS coordinate into a geohash code, we were able to cluster spatially proximate locations into the same or neighboring geohash buckets, facilitating the identification of social encounters and movement patterns. This representation not only significantly reduces the complexity of spatial analysis but also enhances the privacy of the data by abstracting precise locations into more generalized areas.

For the identification of social encounters, our algorithm utilizes the transformation of trajectories into tracks based on the method from our previous work in [3]. Formally, we define track as follows:

Definition 2. Track $\mathcal{T}$ is a sequence of track positions $\{\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3, \dots, \mathcal{P}_m\}$ derived from $\mathcal{S} = \{\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \dots, \mathcal{L}_n\}$, where $1 \leq m \leq n$ and position $\mathcal{P}$ is a tuple $(h, startTime, endTime, id, uid)$ such that:

- h is a geohash code based on location (e.g. *u2s1trm*),
- $startTime$ is a time in seconds from 1.1.1970 (UTC),
- $endTime$ is a time in seconds from 1.1.1970 (UTC),
- id is a unique track identifier,
- uid is a unique user identifier.

Let θ be a conversion function of GPS location to geohash code, $\mathcal{P}_1 = (\theta(lat_1, lon_1), time_1, time_1, id_1, uid_1)$ be the first element of $\mathcal{T}$ such that $\mathcal{L}_1 = (lat_1, lon_1, time_1, id_1, uid_1)$ and $\oplus$ be the successor operator on $\mathcal{T}$ defined as follows:

$$\mathcal{P}_{i \oplus 1} = \begin{cases} (\theta(lat_{i+k}, lon_{i+k}), time_{i+1}, time_{i+k}, id_{i+k}, uid_{i+k}), \\ \quad \text{if } k \text{ is the largest integer } (2 \leq k < n) \text{ such that} \\ \quad \theta(lat_{i+1}, lon_{i+1}) = \theta(lat_{i+2}, lon_{i+2}) = \dots = \theta(lat_{i+k}, lon_{i+k}), \\ (\theta(lat_{i+1}, lon_{i+1}), time_{i+1}, time_{i+1}, id_{i+1}, uid_{i+1}), \\ \quad \text{otherwise,} \end{cases} \quad (1)$$

where $\mathcal{L}_{i+j} = (lat_{i+j}, lon_{i+j}, time_{i+j}, id_{i+j}, uid_{i+j})$ for $i = 1, 2, \dots, n-1$, and either $j = 1$ or $j = k$.

This definition implies that track is a sequence of positions derived from GPS locations, each marked by a unique identifier and time stamps, constructed to represent the movement path of an individual user. This structure enables the detailed analysis of movement patterns and the identification of significant locations within a dataset.

3.3. Discovering social patterns

In the realm of urban mobility and social network analysis, the concept of a meeting plays a pivotal role in understanding how individuals interact within shared spaces. To rigorously analyze these interactions, it is crucial to define what constitutes a meeting in the context of geospatial and temporal data. A meeting is defined as an event where two or more individuals occupy the same geohash location during overlapping time intervals.

Definition 3. A meeting $\mathcal{M}$ within a geohash code h is a set $\{\mathcal{P}_i | \mathcal{P}_i \in \mathcal{T}, uid_i \in U, i \in \mathbb{N}\}$, where U is the set of unique user identifiers, $\mathcal{T}$ is the set of all tracks, and the following conditions hold:

1. $|\mathcal{M}| \geq 2$, indicating the participation of at least two unique users.
2. For all $\mathcal{P}_i, \mathcal{P}_j \in \mathcal{M}$ with $uid_i \neq uid_j$, it must be that $h_i = h_j$, ensuring all positions share the same geohash code.
3. The time intervals $[startTime_i, endTime_i]$ and $[startTime_j, endTime_j]$ intersect for all $\mathcal{P}_i, \mathcal{P}_j \in \mathcal{M}$.

Following the formal definition of meeting, the next step is to explore patterns of continued interaction, or 'followers', within the dataset. This concept delves deeper into the dynamics of mobility, focusing on sequences of meetings that suggest a form of sustained or coordinated movement among individuals. Followers are identified as individuals

or groups who consecutively meet without any intermediate non-meeting locations, suggesting a pattern of shared or coordinated movement through time and space. Below is a formal definition that captures the essence of followers in the context of geospatial-temporal analysis.

Definition 4. Followers are pairs or groups of users $\{uid_i, uid_j, \dots | uid_x \in U\}$ such that for a sequence of meetings $\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_k$, the following conditions hold:

1. There are no track positions for uid_i or uid_j between the meetings $\mathcal{M}_x$ and $\mathcal{M}_{x+1}$ for all $x < k$, i.e., if $\mathcal{P}_y \in \mathcal{M}_x$ and $\mathcal{P}_z \in \mathcal{M}_{x+1}$, then there are no positions $\mathcal{P}_n$ for uid_i or uid_j with a timestamp between $endTime_y$ and $startTime_z$.
2. The users uid_i and uid_j are considered to have a "follower" relationship if their sequence of meetings $\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_k$ occurs without intermediate positions belonging to other geohash codes not part of the meetings.

The definition of followers is the key concept of this paper, providing a fresh way to examine large and complex GPS datasets and uncover social mobility patterns therein.

4. Experimental setup

In this section, we present an overview of the dataset, software and hardware utilized in our paper.

4.1. Software and hardware

In the experimental evaluation for the research paper, the algorithm is implemented in Python 3.10. The experiments are conducted on a desktop PC with a 12-core processor (20 logical processors), and operating at a maximum clock speed of 5 GHz. The desktop is equipped with 32 GB of RAM. Additionally, the experiments benefited from the use of SSD storage. This hardware setup, combined with the capabilities of Python 3.10, created a suitable environment for executing the experimental tests, enabling the algorithm to leverage the available computational resources effectively.

4.2. GPS dataset

The choice of the Geolife dataset [10] is motivated by its unique characteristics and the breadth of opportunities it presents for studying human mobility and social interactions. Developed by Microsoft Research Asia, the Geolife dataset is a rich compilation of GPS trajectories, collected from 182 users over a span of more than five years from April 2007 to August 2012. This dataset not only encompasses a substantial volume of data, over 17,621 trajectories covering about 1.2 million kilometers and 48,000+ hours, but also provides a diverse representation of mobility patterns across different demographics and geographical contexts.

Each trajectory in the dataset is a sequence of time-stamped points, with each point detailing latitude, longitude, and altitude information. These trajectories, recorded by diverse GPS loggers and smartphones, exhibit a wide range of sampling rates. Remarkably, 91 percent of the trajectories offer dense sampling, with points recorded every 1 to 5 seconds or every 5 to 10 meters, providing a detailed depiction of users' movements. The dataset encapsulates a broad spectrum of outdoor movements, from daily routines such as commuting between home and work to leisure and sports activities like shopping, sightseeing, dining, hiking, and cycling. The distribution of logged locations is illustrated in Figure 1, providing a visual representation of the spatial spread and density of the data collected in dataset.

4.3. Data duplicates

The initial phase of our analysis involved a comprehensive cleaning of the Geolife GPS dataset. Cleaning steps included the removal of erroneous data points, such as coordinates outside plausible human mobility ranges or with incorrect timestamps, thereby standardizing the dataset for further analysis.

Detection of duplicates is conducted in two stages: firstly, by filename and secondly, by file content. This two-tiered approach is designed to identify and eliminate redundancies that could skew the analysis of social encounters

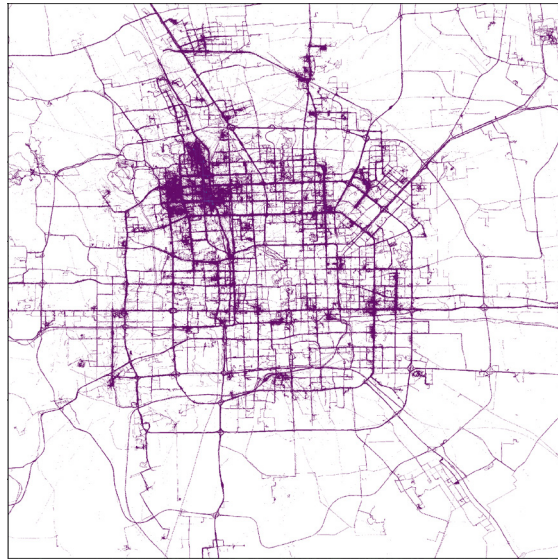


Fig. 1. GPS density locations distribution

and mobility patterns. The initial screening for duplicates involves examining the filenames within the dataset. This step is key for identifying multiple entries submitted by the same device or user, which often resulted from repeated data uploads or system errors. Filenames that match each other are flagged for further inspection. Following the filename-based screening, a more granular examination of file contents is undertaken. This phase involves comparing the timestamps and GPS coordinates within each file in order to identify identical records, indicative of duplicated tracks. Such duplicates not only inflate the volume of data but can also misrepresent the frequency and nature of mobility patterns. By similarity checks, we systematically identify and remove these duplicate entries, ensuring a more accurate dataset for analysis.

An analysis of the dataset reveals varying levels of duplicated files across different users, a summary of which is depicted in Figure 2. This figure highlights the percentage of duplicated tracks for each user, offering insight into the commonness of redundancy within the collected GPS trajectories. Notably, the investigation also uncovers instances of fully copied users' directories of tracks, indicating the presence of fake users within the dataset. This finding further accentuates the critical need for our preprocessing steps, particularly the removal of duplicates and the identification of artificial data contributors, to ensure the integrity and reliability of the mobility patterns analyzed in our study. The disparities in duplication rates among users underscore the importance of data cleaning in studies of human mobility and social interactions, reinforcing the foundational role of accurate data representation in deriving meaningful insights.

5. Experimental evaluation

5.1. Spatial meetings

Analyzing spatial meetings without time constraints sheds light on the distribution of social gatherings within GPS datasets, with findings presented through a sequence of histograms, as seen in Figure 3. The initial analysis, depicted in Figure 3(a), maps the number of persons per meeting against the frequency of such meetings across different geohash codes for the entire dataset. This overview suggests a landscape dominated by numerous smaller gatherings alongside a notable presence of larger groups. To delve deeper, the data is further segmented into more focused histograms. Specifically, Figure 3(b) focuses on meetings involving up to 10 persons, unveiling a predominance of smaller assemblies reflective of typical social interactions or daily routines. Progressing to medium and larger group meetings, the histograms for 10 to 50 persons, shown in Figure 3(c), and for 50 to 100 persons, illustrated in Figure 3(d), reveal an

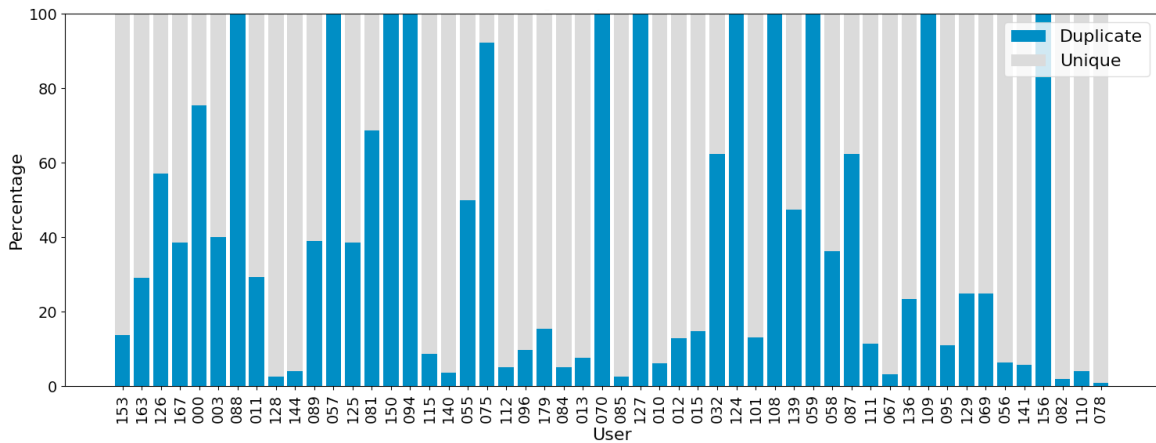


Fig. 2. Duplicate trajectories for various users

incremental decline in the occurrence of such gatherings, underscoring a trend towards less frequent medium to large group meetings within the dataset.

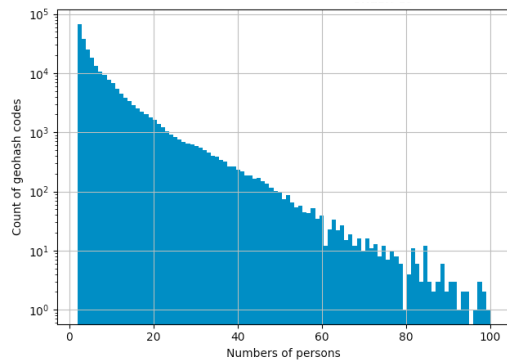
However, the presence of a considerable number of larger gatherings, especially without the filter of time constraints, prompts a reflection on the nature of these meetings. The large counts in certain geohash codes likely correspond to locations such as shopping malls, universities, and workplaces, which inherently attract many people. These areas, characterized by their specific functions and the architectural design to accommodate large numbers of individuals, play an important role in shaping the patterns observed. The distribution of larger meetings in the absence of time constraints thus may not solely reflect social tendencies toward larger gatherings but rather the influence of certain urban spaces that serve as common points of convergence, independent of time. This detailed understanding emphasizes the need to consider the spatial context and the purpose of gatherings, offering a comprehensive perspective on human mobility and spatial interactions within urban environments.

5.2. Spatial-temporal meetings

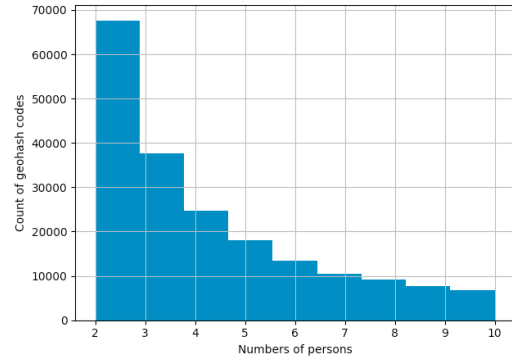
When transitioning from purely spatial to spatial-temporal meetings, time becomes an important factor in defining a meeting. According to our definitions, for a meeting to occur, individuals must not only share the same geohash location but also be present simultaneously, introducing a temporal constraint to the analysis. This addition significantly refines our understanding of social gatherings within the dataset, as evidenced by the patterns observed in the corresponding histogram in Figure 4.

The histogram reveals a stark contrast in the distribution of meetings when temporal constraints are applied, compared to the spatial-only analysis. With the inclusion of time, the largest meeting recorded involves only 5 persons, a scenario that is notably rare, with just one occurrence. The frequency of meetings decreases as the number of participants increases; there are approximately 20 meetings involving 4 persons, around 1,000 meetings with 3 persons, and over 1,000 meetings with just 2 persons. This distribution starkly differs from the spatial analysis without time constraints, where meetings of up to 100 persons were observed.

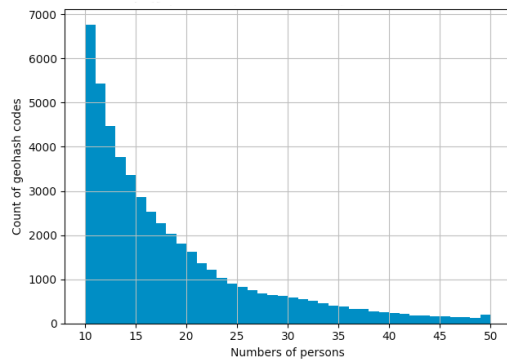
The introduction of the temporal dimension not only narrows the scope of what constitutes a meeting but also highlights the more intimate nature of most social interactions captured in the GPS data. The overwhelming majority of meetings involve small groups of 2 to 3 persons, underscoring the prevalence of brief, possibly incidental or routine, social interactions over larger gatherings. This insight is pivotal for applications that rely on understanding human mobility and interaction patterns, such as urban planning, social network analysis, and public health surveillance, emphasizing the importance of considering both spatial and temporal factors in analyses.



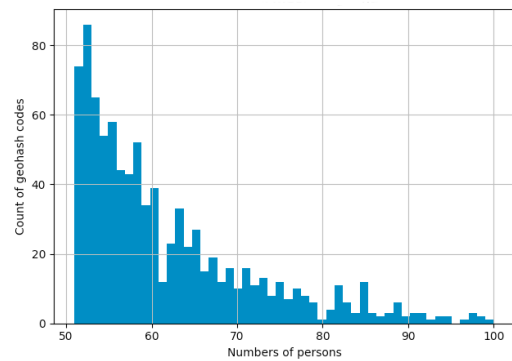
(a) Histogram of spatial meetings - all data



(b) Histogram of spatial meetings - 2 to 10 persons



(c) Histogram of spatial meetings - 10 to 50 persons



(d) Histogram of spatial meetings - 50 to 100 persons

Fig. 3. Histograms of spatial meeting (count of geohash codes in which occurred number of persons at any given time)

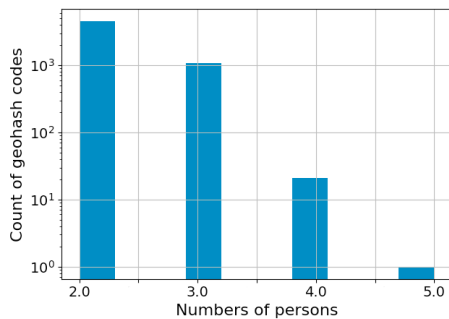


Fig. 4. Histogram of spatial-temporal meetings in geohash codes

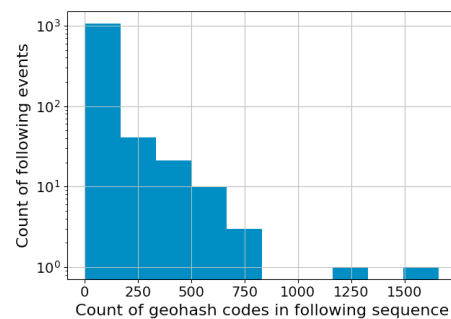


Fig. 5. Histogram of spatial-temporal following events

5.3. Spatial-temporal following events

Based on our definitions incorporating time constraints, the concept of "followers" is further elucidated through an analysis presented in the histogram shown in Figure 5. This figure visualizes the distribution of following events, with the x-axis representing the count of geohash codes in each following sequence and the y-axis showing the number of following events corresponding to these geohash counts. Notably, the shortest following events, involving sequences of just 3 geohash codes, occurred approximately 1,000 times, highlighting a common pattern of brief yet significant

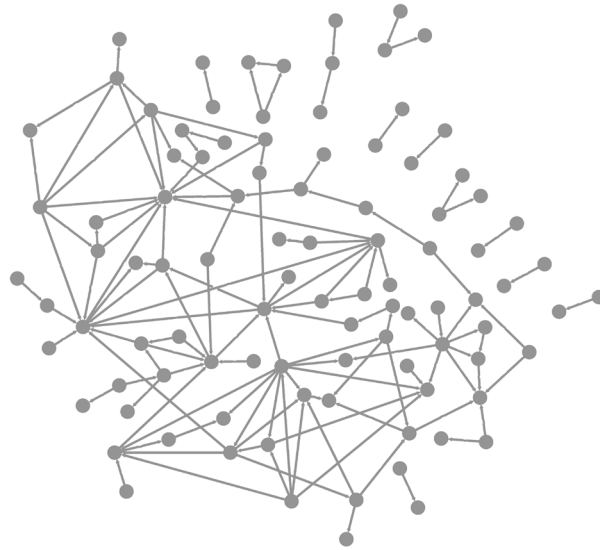


Fig. 6. User following graph

social connections. On the other end of the spectrum, sequences encompassing 500 geohash codes in a single following event get observed 10 times, indicating less frequent but considerably more extended interactions.

Remarkably, the analysis also identifies three exceptional following events that span over 1,000 geohash codes, underscoring rare instances of prolonged and extensive social tracking. These findings, visualized through the histogram, provide a quantitative insight into the nature of followers within the dataset, illustrating a wide range of social dynamics from short-term encounters to long-lasting movements. The presence of such diverse following events sheds light on the intricate patterns of mobility and interaction that characterize the social landscape captured by the GPS dataset, revealing the depth of connections that can be explored through spatial-temporal analysis.

5.4. User following graph

The meetings graph depicted in Figure 6, visually encapsulates the social encounters among users within the dataset. This graph serves as a compelling illustration of interactions captured by spatial-temporal meetings, mapping these interactions across geohash codes. Each *node* in this graph represents a unique user, while the *edges* denote the occurrence of a meeting between two users that are present in the same geohash code and have temporal overlap defined by our meeting criteria as defined in Definition 3. In other words, edges model meetings that do not take place at the same time.

The graph not only highlights the individual connections but also reveals the broader structure of social interactions, illustrating clusters of users who frequently meet, as well as more isolated encounters. This visualization allows for the analysis of social dynamics at a glance, offering insights into how communities might form based on geographic proximity and shared temporal patterns. Furthermore, the density and distribution of connections within the graph reflect the intensity and frequency of social interactions, providing a quantitative measure of social cohesion within the studied area. This representation is invaluable for understanding the complex interplay between spatial mobility and social relationships, offering a unique perspective on human behavior. By mapping the social encounters as a network, we gain a deeper understanding of the underlying patterns that guide social cohesion, community formation, and potentially even the diffusion of information or behaviors across the population.

6. Conclusion

In this research, we delve into the Geolife GPS dataset, which comprises 17,621 trajectories covering approximately 1.2 million kilometers and spans over 48,000 hours across five years. Our approach, adaptable to various GPS

datasets, converts movement trajectories into a series of geohash codes, enhancing the analysis of social behaviors and connections. Remarkably, despite the dataset's size, our analysis, as illustrated in Figure 6, effectively reduces these complex interactions into a simplified representation consisting of only 96 nodes and 131 edges. This reduction to a manageable network graph underscores our method's efficiency in capturing the essence of social connections within such a large dataset.

Unlike other approaches [1, 8, 11], our approach strongly focuses on both spatial-temporal dynamics and GPS locations derived from movements rather than solely on spatial data or GPS locations from social media posts. Our method involves transforming each trajectory into a sequence of geohash codes (tracks) that allows for the efficient extraction of social behavior and connections.

To our best knowledge, we are the first authors to highlight track duplicates and fake "copies" of users that inflate and skew the analysis of social encounters and mobility patterns in Geolife dataset. Our findings after eliminating duplicated, visualized in Figure 6, reveal hidden patterns within the vast expanse of the Geolife dataset, uncovering detailed social encounters, movement patterns, and relationships.

Moreover, we have introduced a web-based platform [4] to showcase these insights across different GPS datasets, demonstrating the broad applicability and potential of our analytical framework. The clarity and accessibility of the depicted social connections in our figures aim to foster a deeper understanding of the Geolife dataset's characteristics, showcasing the intricate social fabric captured through GPS data over the extensive five-year period.

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